

BIOLOGY — GRADE 11 | TAXONOMY II

# Quick Check

## Lesson 3: Pteridophyta and Spermatophyta

*8 multiple-choice questions • choose A, B, C or D*

Sub-Strand 1.3: Taxonomy II | Lesson 3

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**A learner turns over a fern frond and sees rows of small brown dots on the underside. What are these brown dots?**

- A** Flowers that have dried up
- B** Eggs laid by insects
- C** Seeds ready to be planted
- D** Sori, which hold spores for reproduction

**A learner pulls up a maize plant and a bean plant and examines their roots. Which description is correct?**

- A** Both have rhizoids instead of true roots
- B** Maize has fibrous roots and bean has a tap root
- C** Both have a rhizome with fibrous roots
- D** Maize has a tap root and bean has fibrous roots

**A learner examines a leaf that is broad and has veins forming a net pattern. Which plant did the leaf most likely come from?**

**A** Fern

**B** Bean plant

**C** Maize plant

**D** Moss

**A home group is completing its comparison table. Which feature separates a fern (Pteridophyta) from moss (Bryophyta)?**

- A** The fern makes seeds but moss makes spores
- B** The fern has rhizoids only, but moss has true roots
- C** The fern has flowers but moss does not
- D** The fern has true roots and vascular tissue, but moss does not

**A learner says, "Ferns are flowering plants, so they make seeds." Which evidence from the specimens shows this is wrong?**

- A** A fern has no roots, so it cannot make seeds
- B** A fern frond has sori with spores on the underside, and it has no flowers or seeds
- C** A fern frond is green, and only plants with other colours make seeds
- D** A fern frond is smaller than a bean plant, so it makes no seeds

**Both ferns and bean plants have vascular tissue, true roots, stems and leaves. Which feature best separates a fern from a bean plant?**

- A** A fern reproduces by seeds in pods, while a bean plant reproduces by spores
- B** A bean plant has rhizoids, but a fern has true roots
- C** A fern has vascular tissue, but a bean plant does not
- D** A fern reproduces by spores in sori, while a bean plant reproduces by seeds

**A farmer in Kericho and a student in Kisumu both want to place a plant in the same group. Why is checking for spores or seeds better than grouping by size or colour?**

- A** Spores and seeds are always bigger than the plant parts that show colour
- B** Only scientists are allowed to look for spores and seeds
- C** Size and colour never change, but spores and seeds do
- D** Anyone can observe and check spores or seeds in the same way, so people reach the same grouping

In the class flow chart, the first question is "Vascular tissue?" A plant answers "Yes", and the second question is "Seeds?" The plant answers "No, it makes spores". Which group does it belong to?

**A** Pteridophyta (fern)

**B** Bryophyta (moss)

**C** Spermatophyta (maize, bean)

**D** It is not a plant, like the mushroom

# All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).